Hometown Hub – Digital Community Platform

Project Documentation

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Live Website: https://hometown-hub-digital-community.vercel.app/

GitHub: https://github.com/BhavanaJagdale2810/Hometown-Hub-Digital-Community

1. Project Overview

Hometown Hub is a digital community platform designed to help users connect with their local community through posts, communities, events, and user profiles. It provides a central place for members to share updates, discover community activities, and interact with other users. An administrative dashboard supports platform management.

2. Objectives

- Provide a web platform for local community interaction.
- Allow users to register, log in, manage profiles, and publish posts.
- Support community and event discovery and participation.
- Provide an admin interface to monitor and manage users, communities, events, and posts.
- Deploy the application so it can be accessed through a public website.

3. Main Features

- User registration and login with password hashing and token-based authentication.
- User profile and profile-photo support.
- Posts with create, view, like, comment, edit, and delete functionality (as implemented in the application).
- Community listing and community management.
- Event listing and event participation features, including join/leave flows where enabled.
- Notifications for relevant application activity.
- Admin login and dashboard with platform statistics and recent activity.
- Admin management actions for users, communities, events, and posts.
- Forgot-password flow with OTP-based verification, subject to configured email service credentials.

4. Technology Stack

- Frontend: React, JavaScript, Vite, React Router, Axios.
- Backend: Node.js and Express.
- Database: MongoDB with Mongoose; hosted using MongoDB Atlas.
- Authentication: bcrypt password hashing and JSON Web Tokens (JWT).
- File uploads: Multer for supported image uploads.
- Frontend hosting: Vercel.
- Source control: Git and GitHub.

5. System Architecture

The application follows a client-server architecture. The React frontend sends HTTP requests to the Express backend through API endpoints. The backend validates requests, applies authentication and authorization where required, and reads or writes application data in MongoDB. The frontend is deployed on Vercel; the backend API and database must be configured with their production environment settings for live operation.

6. Application Modules

- Public pages: Home, Register, Login, and Forgot Password.
- User pages: Dashboard, Profile, Communities, Events, and Notifications.
- Admin pages: Admin Setup, Admin Login, and Admin Dashboard.

- Backend API: authentication, users, posts, communities, events, notifications, and administrative operations.

7. Live Links

Live website: https://hometown-hub-digital-community.vercel.app/

GitHub repository: https://github.com/BhavanaJagdale2810/Hometown-Hub-Digital-Community

8. Local Setup (Developer Guide)

Prerequisites: Node.js and npm, Git, and a MongoDB connection string. Keep secrets in environment variables; never commit .env files or credentials.

- Clone the repository: git clone https://github.com/BhavanaJagdale2810/Hometown-Hub-Digital-Community.git
- Open the project directory and install frontend dependencies from the frontend folder using npm install.
- Install backend dependencies from the backend folder using npm install.
- Create local environment files using the variable names expected by the backend and frontend configuration. Use your own development credentials and MongoDB URI.
- Start the backend using the package script configured in the backend folder (commonly npm start).
- Start the frontend using npm run dev from the frontend folder.

Exact environment variable names and scripts should be checked in the repository's current configuration before setup. Do not copy production secrets into documentation.

9. Deployment Summary

- Frontend is deployed on Vercel at the live website URL above.
- MongoDB Atlas is used for the hosted database.
- The production frontend must point to the deployed backend API, not localhost.
- Vercel SPA rewrite configuration is used so direct navigation to routes such as /admin-login and /admin-dashboard loads the React application.

Before final submission, verify the deployed backend URL, environment variables, CORS allowed origins, database connectivity, and all public routes. Do not publish secret values.

10. Testing Summary

- Admin Setup successfully created an admin account in the production environment.
- Admin Login and Admin Dashboard opened successfully.
- Dashboard displayed counts for users, communities, events, and posts, plus recent activity.
- Admin View, Block, Unblock, and Delete controls were reported as working during manual testing.
- A final regression pass should also verify user registration/login, post creation, profile, communities, events, notifications, logout, and direct route refresh.

Record the date, test account used, expected result, and actual result for each final test. Use test accounts for destructive actions where possible.

11. Security and Maintenance Notes

- Do not hard-code or publicly share admin passwords, JWT secrets, email credentials, database URIs, or admin setup secrets.
- Use strong unique passwords and store production secrets only in hosting environment settings.
- Restrict or disable public admin-setup functionality after creating the authorized admin, if the backend supports that safely.

- Confirm that every admin API endpoint checks admin authorization on the server; hiding a button in the frontend is not sufficient.
- Use test records when verifying block/delete actions, and ensure the UI confirms irreversible deletion.

12. Known Limitations / Items to Confirm

- Confirm the forgot-password email provider and production email settings before relying on OTP delivery.
- Confirm image upload persistence and file-serving configuration in the deployed environment.
- Confirm that admin setup is protected against unauthorized account creation.
- Confirm that the production backend URL and CORS configuration are correct for all deployed routes.

13. Future Enhancements

- Add stronger audit logs and pagination for administrative lists.
- Improve automated testing for API endpoints and user flows.
- Add monitoring, backups, and a documented recovery process.
- Improve accessibility and responsive behavior based on user feedback.

14. Conclusion

Hometown Hub brings community posts, communities, events, user profiles, notifications, and administrative controls together in one web application. The live admin dashboard has been verified to load and display production data, and core admin management actions have been manually tested. The final step is to complete a full regression check, verify deployment configuration, and attach genuine screenshots of the running application before submission.

Appendix: Screenshot Checklist

- Add a genuine screenshot here: Home page
- Add a genuine screenshot here: Register and Login
- Add a genuine screenshot here: User Dashboard with posts
- Add a genuine screenshot here: Communities page
- Add a genuine screenshot here: Events page
- Add a genuine screenshot here: Admin Login
- Add a genuine screenshot here: Admin Dashboard statistics
- Add a genuine screenshot here: Admin user-management actions